On-device classifier: model comparison

Liquid neural networks (CfC) vs small transformers (Ettin) for the incident-report classifier · measured on a Qualcomm SM8750P dev kit (Snapdragon 8 Elite) · QAIRT/SNPE 2.47 · 10 Oct 2026

> **Recommendation: Ettin-encoder-17m with a trimmed 8k vocabulary**
>
> It is the same size as the best LNN (6.0 MB vs 5.5 MB int8), equal or faster on every processor, the only family that passes the 200 ms line on CPU, loads 5–18× faster, and starts with pretrained English knowledge. The LNN would have to learn the language from ~1,300 conversations. The LNN showed no speed or size advantage.

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model × length builds converted, quantized and ran (42 device runs)

< 3 ms
every model on the Hexagon NPU, ~100× under the 200 ms line

~350 ms
LNN on CPU at 128 tokens; fails 200 ms (Ettin-17m: 8.5 ms)

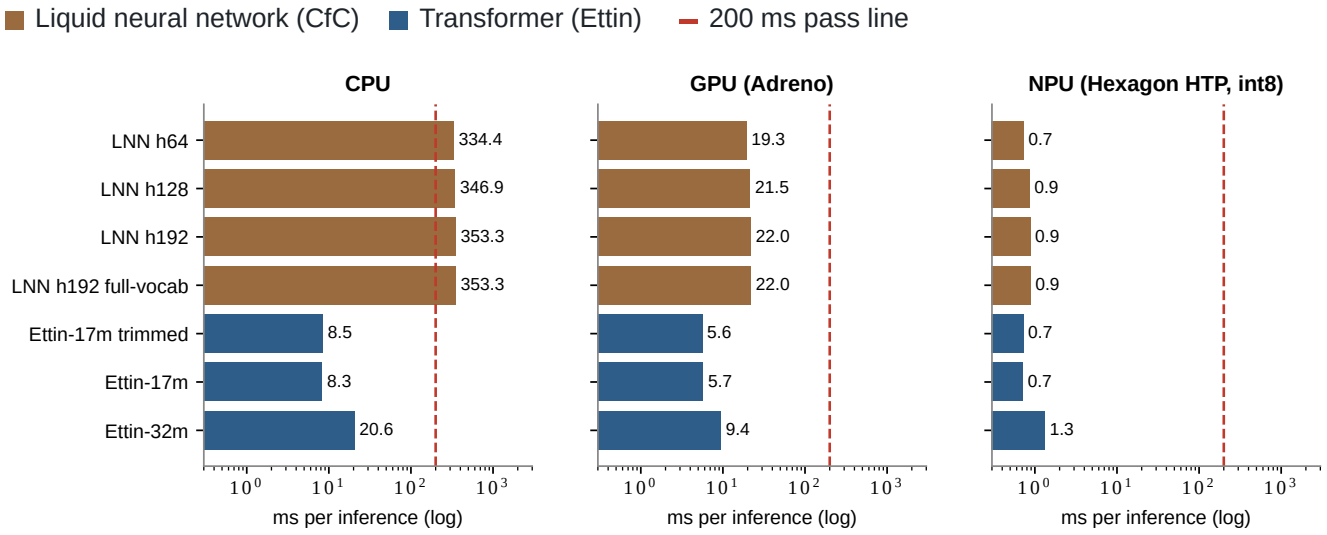
6.0 MB
Ettin-17m trimmed, int8 — same size as the LNNs (5.1–5.5 MB)

Models compared

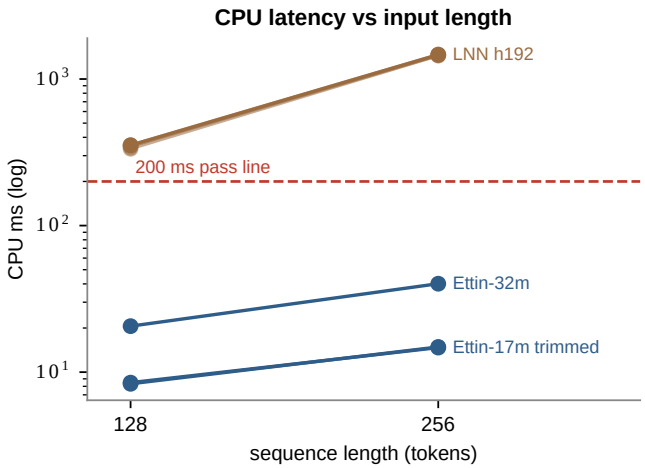
Name	ID	Architecture
LNN h64	`L1h64`	Bidirectional CfC liquid net, hidden 64, 8k vocab
LNN h128	`L1h128`	Bidirectional CfC liquid net, hidden 128, 8k vocab
LNN h192	`L1h192`	Bidirectional CfC liquid net, hidden 192, 8k vocab
LNN h192 full-vocab	`L2h192`	Bidirectional CfC liquid net, hidden 192, full 50k Ettin vocab
Ettin-17m trimmed	`T1t`	Ettin-encoder-17m (7 layers, d=256), vocab trimmed to 8k
Ettin-17m	`T1`	Ettin-encoder-17m (7 layers, d=256), full 50k vocab
Ettin-32m	`T2`	Ettin-encoder-32m (10 layers, d=384), full 50k vocab

All models take the same input (Ettin tokenizer, int32 `input_ids` + `attention_mask` of shape [1, L]) and have the same three heads: incident type (30 classes = 28 types + undetermined + none), per-token relevance (2) and per-token form fields (60, multi-label). ModernBERT-base was excluded by decision.

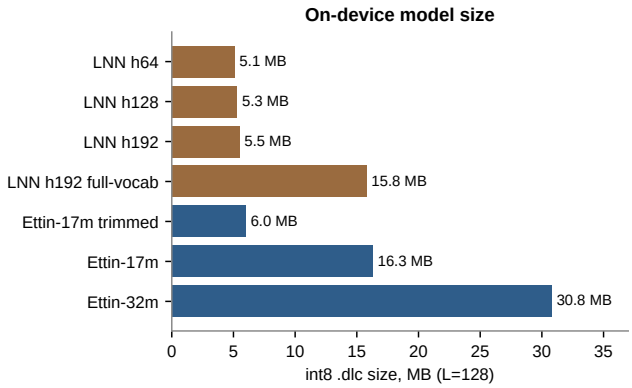
Latency on device



Average of 50 real conversations, 128-token input, burst performance mode. Log scale.



The unrolled LNN grows ~4× on CPU when the input doubles; Ettin grows ~2×.



The word-embedding table is most of every model. Trimming it brings Ettin-17m down to LNN size.

Full results

Model	L	Params	ONNX ops	DLC fp32 MB	DLC int8 MB	Convert s	CPU ms	GPU ms	NPU ms	NPU load s
LNN h64	128	2.18M	5,648	11.2	5.1	72	**334.4**	19.3	0.7	1.8
LNN h64	256	2.18M	11,280	14.1	8.1	145	**1446.2**	37.2	1.2	6.0
LNN h128	128	2.35M	5,648	11.8	5.3	73	**346.9**	21.5	0.9	1.8
LNN h128	256	2.35M	11,280	14.7	8.2	146	**1450.8**	41.6	1.5	5.9
LNN h192	128	2.60M	5,648	12.8	5.5	72	**353.3**	22.0	0.9	1.7
LNN h192	256	2.60M	11,280	15.7	8.5	146	**1466.8**	43.8	1.7	5.9
LNN h192 full-vocab	128	13.40M	5,648	54.0	15.8	76	**353.3**	22.0	0.9	1.7
LNN h192 full-vocab	256	13.40M	11,280	56.9	18.8	145	**1466.7**	43.5	1.8	5.9
Ettin-17m trimmed	**128**	**6.02M**	**325**	**23.3**	**6.0**	**9**	**8.5**	**5.6**	**0.7**	**0.3**
Ettin-17m trimmed	**256**	**6.02M**	**325**	**23.4**	**6.1**	**9**	**14.7**	**7.9**	**1.4**	**0.3**
Ettin-17m	128	16.82M	325	64.4	16.3	9	8.3	5.7	0.7	0.5
Ettin-17m	256	16.82M	325	64.6	16.4	9	14.9	8.2	1.4	0.5
Ettin-32m	128	31.92M	457	122.1	30.8	13	20.6	9.4	1.3	0.8
Ettin-32m	256	31.92M	457	122.2	30.8	12	40.1	13.4	2.5	0.8

CPU and GPU run the fp32 model; NPU runs the int8 model (snpe-dlc-quant, calibrated on the same 50 conversations). Red = fails the 200 ms line. "NPU load" is the first-time init on the Hexagon HTP. CPU fallback was disabled, so every successful NPU/GPU run executed all layers on that processor.

Findings

Speed

- **NPU:** every model takes 0.7–2.5 ms. Speed cannot choose between them on the Hexagon HTP.
- **GPU:** all pass (5.6–44 ms); Ettin is 2–5× faster than the LNNs.
- **CPU:** the LNNs fail (334–353 ms at 128 tokens, ~1.45 s at 256). Ettin-17m passes at 8–15 ms, Ettin-32m at 21–40 ms. The LNN has to be unrolled into a fixed graph of 5,600–11,300 tiny sequential ops, and the CPU runtime pays overhead for each. A transformer is 325–457 large, parallel ops.
- **Hidden size barely matters** for the LNN (h64 vs h192: 334 vs 353 ms CPU), so per-op overhead, not maths, dominates.

Size and load time

- The embedding table dominates: LNN h192 is 5.5 MB with an 8k vocab and 15.8 MB with the full 50k vocab.
- Same 8k vocab: Ettin-17m 6.0 MB vs LNN 5.1–5.5 MB int8 — effectively equal.

- The LNN's file grows with input length (5 → 8 MB from 128 → 256 tokens) because of unrolling; Ettin's does not.
- First load on the NPU: LNN 1.7–6.0 s, Ettin 0.3–0.8 s.

Input length

The existing labelled conversations are 43 tokens (median) and 71 tokens at most, so a 128-token input covers them all. 256 only matters if the new multi-turn data runs much longer.

Method

- **Device:** Qualcomm dev kit "Sun", SoC SM8750P (Snapdragon 8 Elite), Hexagon v79 HTP, Android 15. The Snapdragon 695 test phone was not benchmarked.
- **Pipeline:** PyTorch → ONNX (opset 17, static shapes, onnxsim; onnxruntime matched PyTorch within 1.8e-6) → `qairt-converter` → `snpe-dlc-quant` (int8) → `snpe-net-run` with `--perf_profile burst --profiling_level basic` → `snpe-diagview`.
- **Inputs:** 50 real conversations from `data/*.jsonl`, tokenized with the Ettin tokenizer, padded to L.
- **Weights are random.** Ettin was built from its published config. Speed and size do not depend on weight values.

Limits

- Times are model execution only: no tokenization or app overhead.
- Accuracy after int8 quantization is untested; measure it once the model is trained.
- An LNN written as a plain loop in app code (outside the Qualcomm SDK) might be much faster on CPU. Not measured.
- NPU load time could drop with an offline-prepared graph cache (`snpe-dlc-graph-prepare`). Not measured.
- Trimming Ettin's vocabulary means keeping only the pretrained embedding rows for tokens the form pack uses (plus common English), with an unknown-token fallback.

Source: `bench/` in the megathon repo — `RESULTS.md`, `results.json`, per-run logs in `bench/out/<model>_L<len>/`. Regenerate with `python bench/report/make_report.py`.